

TEST PLAN

Language Learning Hub- Flashcard CRUD, AI Generation, Sharing & Export

Testing Portfolio Project

Github: [LLH-Functional-and-API-Testing-Project](#)

Github Web Project: [Language-learning-hub-for-quizz](#)

Url: [link web](#)

Requirement Basis: BRD v1.9 (BR-02), FRD-DOC v0.5 (FR-DOC-009–019), Phase 2 CRs (CR-DOC-001, CR-DOC-002, CR-CROSS-001), Production Hotfixes (DEF-AI-01–04, DEF-UI-01–02, DEF-SAVE-01), Requirement Analysis.

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1. Analyze the Product

1.1 Target Actors

Actor	Description
Learner (owner)	Authenticated, Active account. Has full CRUD rights over their own Flashcards, Decks, and Folders.
Learner (non-owner)	Authenticated, Active account. May not access or mutate another Learner's resources.
Public User	Unauthenticated visitor. View-only access to an active shared Deck URL; no other actions permitted.
AI Service (Grok)	External provider invoked by the system to generate temporary Flashcard preview content.
System	Enforces session validity, Active account status, ownership, public-link state, and downstream data consistency.

1.2 Test Basis

Document	Version / Status
Requirement Analysis	requirements_spec_DOC_flashcard-sharing-export.md - finalized, all Ambiguities resolved
BRD	v1.9 (BR-02)
FRD-DOC	v0.5 (FR-DOC-009–019)
Phase 2 Change Requests	CR-DOC-001, CR-DOC-002, CR-CROSS-001
Production Hotfixes	DEF-AI-01–04, DEF-UI-01–02, DEF-SAVE-01
Data Dictionary	DATA-DICTIONARY.md

1.3 Scope

In Scope

- View Flashcards in an owned Deck, including empty-list state.

- Manually create, edit, and permanently delete an owned Flashcard (FR-DOC-009 to FR-DOC-012).
- Validate Flashcard content (required fields, character limits, language placement, normalized duplicate term, immutable Deck association).
- Verify initial SRS state on save and SRS removal on delete.
- Submit vocabulary list to AI, validate parsed/deduped input, and verify temporary AI preview generation (FR-DOC-013 to FR-DOC-014).
- Review, edit, and save selected AI-generated Flashcards to an existing or new Deck (FR-DOC-015).
- Enable/disable public sharing link for an owned Deck; verify token creation, revocation, and reuse prevention (FR-DOC-016 to FR-DOC-018).
- View an active shared Deck as a Public User; verify data minimization and read-only enforcement (FR-DOC-017).
- Export a non-empty owned Deck to .xlsx; verify file content, column order, filename format (UTC+7), and non-mutation of application data (FR-DOC-019).
- Cross-module impact checks: verify that Flashcard CRUD, AI save, sharing, and export do not advance profiles.last_learning_activity_at or create notification rows.
- RBAC and ownership enforcement at both UI and API layers.
- API-level validation for Flashcard CRUD, AI Preview + Save, Sharing, and Export endpoints.
- Database-level data verification for SRS initialization, deletion cleanup, and sharing state.

Out of Scope

- Folder and Deck CRUD (FR-DOC-001 to FR-DOC-008).
- Moving Flashcards between Decks.
- Editing, importing, copying, or learning from a shared public Deck.
- Non-functional testing (UI, Usability , Compatibility, Performance, Security)
- AI content accuracy guarantees; internal token algorithm and database schema design.
- Persistent AI metadata; AI-generated content exists only until saved or discarded.

2. Define the Test Approach

2.1 Approach Summary

Dimension	Applied Approach
Test level	System test

Test types	Functional testing (primary); cross-module checks; database data verification
Execution mode	Manual
Test-case approach	Positive, Negative, Edge case using black box testing
Database verification	Direct DB query to confirm SRS initialization, deletion, sharing state, and absence of activity/notification records
Cross-module checks	Verify the side effect of tested function to others
Exploratory testing	Supplementary; used for AI generation edge cases and sharing/export corner cases after scripted coverage

2.2 Main Test Areas

#	Test Area
TA-01	Flashcard View - owned list, empty-list state, cross-owner denial, missing Deck
TA-02	Flashcard Create - required fields, trim/limit, language placement, normalized duplicate, SRS init, RBAC
TA-03	Flashcard Edit - content update, immutable Deck association, learning-state preservation, validation rollback, RBAC
TA-04	Flashcard Delete - permanent deletion without confirmation, SRS removal, downstream exclusion (LEARN/DASH/share/export)
TA-05	AI Input Validation - parsing (comma/newline/mixed), trim, dedup, 50-item/3,000-char limits, missing/unsupported targetLanguage
TA-06	AI Preview Generation - field validation, language mapping, Sino-Vietnamese rules, failedItems reporting, timeout/partial/failed outcomes
TA-07	AI Preview Review & Save - editable preview, duplicate pre-check on UI, save to existing/new Deck, SRS init, discard boundary, cross-module non-impact
TA-08	Sharing Enable - token/URL activation, idempotency (re-enable shows same URL), ownership enforcement

TA-09	Shared Deck Public View - active-token validation, data minimization, view-only, unavailable state for invalid/disabled/deleted link
TA-10	Sharing Disable & Revocation - immediate revocation, fresh-request unavailability, new token on re-enable, old-token invalidation
TA-11	Export - .xlsx content, column order, filename UTC+7, multiline preservation, empty/unauthorized denial, non-mutation of application data, export failure handling
TA-12	RBAC & Session - ownership enforcement for all actions at UI and API layer; deactivated Learner session; Admin/non-owner/Public User denial

2.3 Test Design Techniques

Technique	Applied To
Equivalence Partitioning	Field validation (valid/invalid/empty/over-limit values for term, meaning, exampleSentence); targetLanguage (Japanese / Chinese / missing / unsupported)
Boundary Value Analysis	term max 500 chars, meaning max 1,000 chars, exampleSentence max 500 chars; AI input max 50 items and 3,000 characters
Decision Table Testing	Language placement rules (Japanese Kanji → Sino-Vietnamese required; Japanese Hiragana/Katakana → Sino-Vietnamese null; Chinese Hanzi → Sino-Vietnamese required)
State-Transition Testing	Flashcard lifecycle (Draft → SavedNew → Updated → Deleted); Sharing-link lifecycle (Disabled ↔ Active → Deleted); AI generation lifecycle (InputDraft → Pending → PreviewCompleted/PreviewPartial/Failed → Saved/Discarded)
Error Guessing	Concurrent duplicate-term submissions; rapid re-enable/disable of sharing; export of an empty Deck
Exploratory Testing	AI edge cases; public-page browser Back behavior after revocation (accepted per AMB-01)
Pairwise Testing	AI input format combinations (comma-separated / newline / mixed) × language selection (Japanese / Chinese)

2.4 Test Priority

Impact	Meaning for this Project
High	Feature is a core user journey or security/data-integrity control; failure blocks a key function or exposes a data breach / ownership violation. Execute first and require full evidence before exit.
Medium	Feature supports a significant user journey or adds important guardrails; failure causes an incomplete or degraded experience but not a data breach. Execute after all High items pass.
Low	Feature is a convenience or cosmetic quality item; failure does not block any primary flow. Execute after Medium items; acceptable to defer in a time-constrained cycle.

Risk-based prioritization factors:

- **High:** RBAC/ownership enforcement (RISK-01); sharing revocation and token management (RISK-02); public data leakage (RISK-03); data integrity after deletion (RISK-04); duplicate-term bypass including Chinese legacy (RISK-05).
- **Medium:** AI generation lifecycle correctness (RISK-08, RISK-09); AI save boundary and cross-module non-impact (TA-07, TA-13); export content/privacy correctness (RISK-06); Flashcard create/edit validation.
- **Low:** Export failure handling and retry (AMB-04 resolved); empty-state messaging; idempotent sharing re-enable; export filename format.

2.5 Test Data Approach

Data categories required

Category	Description
Valid Flashcard data	Japanese and Chinese term/meaning/exampleSentence within limits; Kanji + Sino-Vietnamese; Hiragana-only + null Sino-Vietnamese; Chinese Hanzi + Sino-Vietnamese + optional Pinyin
Invalid/Boundary data	Blank fields after trim; fields at exactly max length (500, 1,000 chars); fields 1 character over max; normalized duplicates (whitespace, case, Unicode variants)
AI vocabulary lists	1-item valid list; 50-item boundary list; 51-item over-limit list; mixed comma/newline/blank lines; all-blank input; items mismatched with selected language; 3,000-char boundary; 3,001-char over-limit

Cross-owner data	Two Learner accounts with distinct Decks; cross-owner access attempts using Learner B credentials on Learner A resources
Export target Deck	Non-empty owned Deck with known Flashcard count, multiline field content, and at least one field containing special characters

3. Define Test Criteria

3.1 Entry Criteria

- Finalized, approved requirement analysis document is available and all Ambiguities are resolved.
- Production application is reachable and stable at the target URL.
- At least two distinct Learner test accounts with Active status are available and accessible on Production.
- Direct database query access (read-only) is confirmed for SRS state and notification verification.
- Test case documents (Excel/Docs) are drafted and reviewed before execution begins.
- Test data naming conventions and cleanup procedures are documented and agreed.

3.2 Test Result Criteria

Status	Definition
PASS	All steps execute as expected; actual results match expected results; no material unexpected side effects are observed.
FAIL	Actual result deviates from the expected result in a way that is supported by the requirement; deviation is recorded with evidence.
BLOCKED	Test case cannot be executed due to a missing dependency, environment issue. Reason must be recorded.
NOT RUN	Test case was not executed within this cycle; reason must be recorded.

3.3 Exit Criteria

- All High-priority and Medium-priority test areas have been executed and results recorded.
- No open High-severity defects that block a core user journey remain unaddressed (either fixed and retested, or explicitly accepted with residual-risk sign-off).

- All FAIL and BLOCKED results have been recorded.
- A Test Summary Report has been produced, listing execution metrics, open defects, residual risks, and test coverage assessment.

4. Resource Planning

Tool	Purpose
Google Chrome	Windows 11
Excel	Test case authoring, execution recording, defect log
Google Docs	Test plan, test summary report, requirement analysis reference
GitHub Issues	repository: LLH-Mannual-API-testing-project)
Database client	Direct read-only query against Production DB
Screen capture tool	Evidence capture (screenshots) for FAIL records

5. Plan the Test Environment

5.1 Test Environment

Item	Configuration
Application	Language Learning Hub - Production build
Environment URL	https://language-learning-hub-for-quizz-fro.vercel.app
Module under test	DOC - Flashcard CRUD, AI Generation, Sharing, Export (FR-DOC-009 to FR-DOC-019)
Operating System	Windows 11
Browser	Google Chrome
Test accounts	Learner A (owner), Learner B (non-owner/cross-owner)
External integration	Groq AI Service - live, shared free tier
Database access	Read-only query access to Production DB
Network	Standard internet connection; no VPN or proxy required

5.2 Environment Limitations

Limitation	Impact
Production is the only test environment	Test data is mixed with real user data; cleanup failures leave residual records.
Groq AI Service is a shared free tier	Rate-limit (429) or service outage can block AI test cases unpredictably.
Groq AI content is non-deterministic	Generated field values vary per call, making exact-match assertions on AI content unreliable.
No staging or rollback capability	Defects found in Production cannot be isolated; test data pollution is permanent until cleaned up.

6. Schedule and Estimation

Activity	Estimated Effort
Test planning (this document)	0.5 day
Test case design	3 days
Execution	2,5 days
Defect report	1 day
Test Summary Report	0.5 day
Total estimated effort	~7,5 days

7. Determine Test Deliverables

- Test Plan
- Requirement Analysis
- Test Case Workbook + Execution Records (Excel) for Functional testing
- Test Data (Excel)
- Bugs Reports (Docs)
- Defect Reports (GitHub Issues)
- Test Summary Report